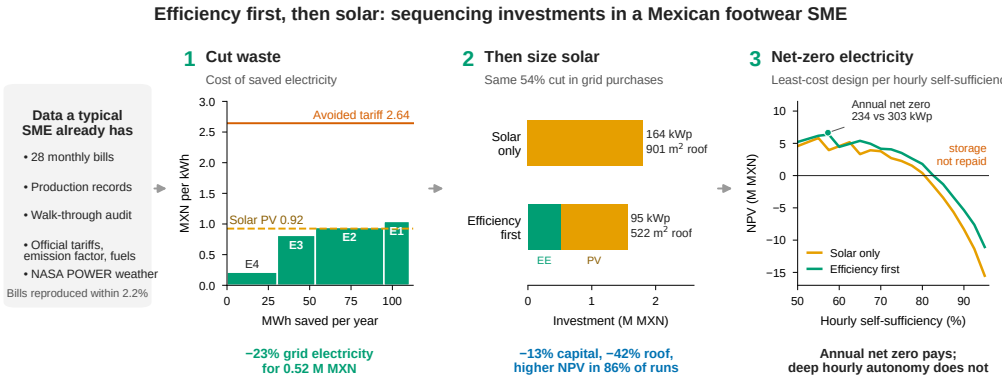
Graphical Abstract

Efficiency first, then solar: cost, emissions and energy-security trade-offs of demand- and supply-side options in a Mexican footwear SME

Yeniba Alberú Argüeso, Ivan Aguado Guerra, Francisco Acosta Sosa, Samuel Antonio Rosas-Meléndez, Iván García Kerdan



Highlights

Efficiency first, then solar: cost, emissions and energy-security trade-offs of demand- and supply-side options in a Mexican footwear SME

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- Bill-validated tariff model shows GDMTO demand charges are driven by energy use
- Demand-side measures cost less per kWh than the tariff they avoid
- Efficiency first cuts PV need by 42% for the same 54% grid reduction
- Efficiency savings are robust to net-metering rules; PV value is not
- Annual net-zero electricity pays off; storage for 80% autarky does not yet

Efficiency first, then solar: cost, emissions and energy-security trade-offs of demand- and supply-side options in a Mexican footwear SME

Yeniba Alberú Argüeso[a], Ivan Aguado Guerra[a], Francisco Acosta Sosa[a], Samuel Antonio Rosas-Meléndez[a], Iván García Kerdan[a,*]

[a] *Tecnologico de Monterrey, School of Engineering and Sciences, Mexico*

Abstract

Energy efficiency first (EE1st) asks decision-makers to treat demand-side resources as alternatives to supply investments, but evidence for small and medium-sized manufacturers in emerging economies is scarce. We apply the principle to a safety-footwear plant in Pachuca, Mexico, that buys 460.8 MWh of grid electricity per year under the medium-voltage GDMTO tariff. Twenty-eight months of bills, production records, official tariffs and an audit end-use estimate feed a transparent model that reproduces reported prices with a mean absolute error of 2.2%. Because billed demand is driven by monthly energy, each avoided kWh is worth 2.64 MXN. Base-load control, insulation of heated equipment, servo motors and a compressed-air package save 23% of electricity at 0.21–1.04 MXN per kWh, below the avoided tariff in more than 99.9% of 5,000 Monte Carlo runs. Rooftop photovoltaics are also profitable, but sequencing matters: efficiency first reaches the same 54% cut in grid purchases with 94.9 instead of 163.8 kWp, 13% less capital and a higher net present value in 86% of runs. It also lowers the energy-implied peak by 32 kW and keeps its value if photovoltaics lose demand-charge credit, which would cut the array's value by 69%. Annual net-zero electricity is profitable with 234 kWp after efficiency (69 kWp less than without it), whereas 80% hourly self-sufficiency needs 300 kWh of batteries that current tariffs do not repay; EnergyPlus reproduces the hourly photovoltaic and battery flows. For SMEs, efficiency first works best as a sequencing rule: cut waste, then size

*Corresponding author.
Email address: i.garciakerdan@tec.mx (Iván García Kerdan)

supply.

Keywords: Energy efficiency first, Industrial SMEs, Energy security, Cost of conserved energy, Distributed photovoltaics, Electricity tariffs, Mexico

1. Introduction

Decarbonising industry while protecting firms from volatile energy markets has become a shared objective of climate and energy-security policy. The European Union has made energy efficiency first (EE1st) a legal principle: decision-makers must consider cost-effective demand-side alternatives before investing in supply, networks or markets (European Parliament and Council of the European Union, 2023; European Commission, 2021, 2024). Global progress nonetheless remains slow. Energy intensity improved by about 1% in 2024, far below the doubling pledged at COP28, and governments increasingly justify efficiency policy on grounds of security, affordability and competitiveness (International Energy Agency, 2024, 2025). EE1st is therefore more than an accounting rule for emissions. It is a decision principle about which resource is procured first, and in what amount, when demand reduction and supply expansion compete for the same capital.

Most quantitative work on EE1st concerns public decisions in Europe, such as building-sector pathways, power-sector incentives and the design of policy instruments (Mandel et al., 2022; Pató and Mandel, 2022; Mandel et al., 2023; Mandel and Pató, 2024). Far less is known about how the principle performs inside individual firms, particularly small and medium-sized enterprises (SMEs) in emerging economies. There, investment decisions are made by owners with limited technical staff, short payback expectations, constrained credit and tariffs that were not designed with efficiency in mind (Cagno et al., 2013; Trianni et al., 2017; Kalantzis and Revoltella, 2019; Zhang et al., 2025; Herce et al., 2024). In such firms the dominant "decarbonisation" proposal is often rooftop photovoltaics (PV). PV is visible, financeable and supported by distributed-generation rules, whereas the cheaper but less tangible demand-side measures are postponed or never quantified (Fleiter et al., 2012; Thollander et al., 2025).

Mexico illustrates why this sequencing matters for energy security as well as cost. In 2024 combined-cycle and gas-turbine plants supplied about two-thirds of the electricity injected into the national grid (Secretaría de Energía, 2025). US pipeline imports covered roughly three-quarters of national natural-

gas consumption (U.S. Energy Information Administration, 2025). The official grid emission factor rose from 0.435 to 0.444 $tCO_2e\ MWh^{-1}$ between 2022 and 2024 (Comisión Reguladora de Energía, 2023, 2024; Secretaría de Medio Ambiente y Recursos Naturales, 2025). Every industrial kWh therefore carries an embedded exposure to imported fossil fuel. The 2025 electricity-sector law also raised the threshold for exempt distributed generation to 0.7 MW (Cámara de Diputados, 2025), which makes on-site PV more attractive for SMEs. For a medium-voltage industrial customer, the question is no longer whether PV pays, but whether efficiency should come first, and what the firm and the grid lose if it does not.

This paper answers that question for a safety-footwear plant in Pachuca, Hidalgo. The work builds on an energy review and measurement-and-verification (M&V) plan prepared during a graduate energy-management project, consistent with ISO 50001 and the national implementation manual (ISO, 2018; CONUEE and GIZ, 2022). We re-estimate the energy baseline from the original records and build a tariff model from official charges, validated against the plant's bills. We then compare demand-side measures and supply-side options on common indicators: energy, emissions, cost of conserved energy, net present value (NPV) and a set of plant-level security and resilience metrics. The analysis is an *ex-ante* evaluation study. It does not observe post-retrofit performance, but it is built on measured bills and official tariffs and quantifies uncertainty explicitly, so that its limits are visible. All data and code are released for reuse.

The rest of the paper is organised as follows. Section 2 reviews the literature and states the gap. Section 3 describes the case, data and model. Section 4 presents the results. Section 5 discusses their implications and limitations, and Section 6 concludes.

2. Literature review

2.1. The energy efficiency first principle

EE1st originates in least-cost and integrated resource planning, in which demand-side resources compete on equal terms with generation and network capacity. Mandel et al. (2022) conceptualise it as a decision principle with three ingredients. These are an equal-footing comparison of demand- and supply-side resources, a societal perspective that includes multiple benefits, and a dynamic view of how decisions interact over time. The EU recast Energy Efficiency Directive turned the principle into an obligation for large

planning, policy and investment decisions (European Parliament and Council of the European Union, 2023). Commission guidance recommends life-cycle cost-benefit analysis that includes carbon prices and non-energy benefits (European Commission, 2021, 2024). Below the investment thresholds of the Directive, which include almost all decisions taken by SMEs, the principle is encouraged rather than mandated (European Commission, 2024). Modelling studies show that efficiency-led pathways can lower the total cost of reaching net zero in buildings by avoiding supply and network investment (Mandel et al., 2023). In the power sector, network-company incentives determine whether demand-side resources are procured at all (Pató and Mandel, 2022). Recent work classifies the policy instruments that can operationalise the principle and highlights an implementation gap between legal recognition and actual decisions (Mandel and Pató, 2024).

Two features of this literature limit its transferability to firms. First, applications are dominated by system-level or sectoral models in high-income economies, where the comparison is between policy pathways rather than between assets that a single owner can buy. Second, the indicators used to judge EE1st, such as system cost and avoided capacity, rarely include the variables that drive industrial decisions. These include tariff structure, roof-space limits, regulatory risk attached to distributed generation and exposure to energy-price shocks.

2.2. Energy efficiency in industrial SMEs

Industrial SMEs face well-documented barriers: limited information and technical skills, competing investment priorities, split incentives and high perceived risk (Cagno et al., 2013; Trianni et al., 2017). Energy audits raise awareness, but adoption of recommended measures remains partial. Low-cost organisational measures and measures with short payback are adopted most often (Fleiter et al., 2012; Kalantzis and Revoltella, 2019). Programme design, energy-efficiency networks and access to finance are important levers (Johansson et al., 2019; Jalo et al., 2021; Herce et al., 2024; Zhang et al., 2025). Energy management is increasingly understood as a socio-technical practice in which routines, roles and organisational learning matter as much as technology (Thollander et al., 2025). International uptake of ISO 50001 has been promoted as a scalable route to such practices (McKane et al., 2017; Moreno et al., 2026). Non-energy benefits, such as productivity, maintenance, noise and thermal comfort, often equal or exceed the value of the energy saved,

yet firms seldom include them in investment calculations (Worrell et al., 2003; Nehler and Rasmussen, 2016; Nehler, 2018).

For the measures considered here the evidence base is uneven. Compressed-air systems are among the best documented. Leaks commonly waste 20–30% of compressor output in poorly maintained plants, and every 2 psi of excess pressure adds roughly 1% to compressor energy (U.S. Department of Energy and Compressed Air Challenge, 2016; Saidur et al., 2010; Dindorf, 2012). Replacing clutch motors on industrial sewing machines with servo motors is reported to cut motor energy by 65–70% (Nemeša, 2023). Idle and base loads frequently dominate specific energy use at low throughput, so operational switch-off strategies can deliver large savings at negligible cost (Mouzon et al., 2007; Kara and Li, 2011; Gutowski et al., 2009). Footwear-specific evidence focuses on product carbon footprints. It shows that factory electricity is a material contributor to the footprint and that results depend strongly on metering resolution (Milà et al., 1998; Cheah et al., 2013; Zhang et al., 2026). Production scheduling has been shown to reduce energy by about 5% (Sudhakar and Muthucumaraswamy, 2026). Peer-reviewed quantification of insulation savings for heated barrels and platens is scarce. Practitioner guidance reports short paybacks (Kent, 2018).

2.3. Comparing demand- and supply-side options

The cost of conserved energy (CCE), the annualised cost of a measure divided by the energy it saves, provides a common metric for ranking efficiency against energy prices or supply costs (Meier et al., 1982; Tesema and Worrell, 2015). Marginal abatement cost curves extend the idea to emissions. They are criticised for ignoring interactions between measures, uncertainty, non-financial costs and path dependency (Kesicki and Ekins, 2012; Huang et al., 2016). Supply-side options for industrial SMEs include on-site PV, whose levelised cost has fallen sharply (International Renewable Energy Agency, 2025). Mexican case studies report attractive returns under net metering (Delfín-Portela et al., 2023; Linvill and Brutkoski, 2017). A further option is electrification of process heat, where heat pumps can supply low-temperature heat with coefficients of performance (COP) well above one (Arpagaus et al., 2018; Schlosser et al., 2020; Madeddu et al., 2020; International Energy Agency, 2022). Because savings are uncertain and energy prices volatile, risk-aware appraisal methods such as Monte Carlo simulation are recommended for efficiency investments (Mills et al., 2006; Jackson, 2010). Measured savings should be verified ex post through protocols such as IPMVP (Efficiency

Valuation Organization, 2022). Firm-level evidence shows that energy-price uncertainty also shapes manufacturers' investment efficiency (Anh et al., 2024).

2.4. Research gap and contribution

Taken together, the literature offers a well-defined principle, mature appraisal metrics and abundant evidence on individual measures. It offers little evidence on how EE1st changes concrete investment decisions of SMEs operating under emerging-economy tariffs and fossil-import dependence. Three gaps stand out. First, plant-level studies typically appraise efficiency measures or PV in isolation, so they cannot show how efficiency changes the size, cost and roof requirement of the supply option that follows. Second, the tariff structure is usually reduced to an average price. That hides whether demand charges reward energy savings, peak reduction or neither, and therefore how robust each option is to regulatory change. Third, security and resilience are discussed qualitatively at the firm level, even though they are a central motivation for the principle.

This study addresses these gaps with a reproducible, bill-validated appraisal of a Mexican footwear SME. Its contribution is fivefold. (i) It places demand-side measures, rooftop PV and fuel switching on common indicators of energy, emissions, CCE, NPV and security, using official tariffs, prices and emission factors. (ii) It quantifies EE1st as a sequencing decision, comparing supply-only and efficiency-first pathways that reach the same reduction in grid purchases. (iii) It introduces plant-level security indicators: imported gas embodied in grid purchases, bill exposure to an energy-charge shock, regulatory stress on distributed-generation compensation and proximity to the tariff-class threshold. (iv) It tests whether the plant can reach net-zero electricity with renewables and storage, distinguishing annual netting from hourly self-sufficiency. (v) It releases the code, public input data and derived results, together with an open toolkit that generalises the method so that other SMEs can repeat the assessment from their own monthly bills.

3. Methodology

3.1. Case study and data

The plant produces safety footwear in the Pachuca de Soto area, Hidalgo (city coordinates 20.10°N, 98.76°W, about 2,340 m above sea level). Processes include cutting, stitching, lasting, rubber vulcanising and direct sole injection



Figure 1: Location of the plant (Pachuca de Soto, Hidalgo). Boundaries: Natural Earth.

moulding. Production is flexible. The plant normally operates from 08:00 to 18:00 on weekdays and from 08:00 to 13:00 on Saturdays, and it ran 24-hour shifts during a production surge in November 2025–January 2026. It is supplied at medium voltage under the CFE GDMTO tariff (ordinary large demand, medium voltage), Centro Oriente division. Offices and warehouses share the meter but were outside the scope of the audit. The company and its exact location are not disclosed. Its billing, production and audit records are confidential and are reported here only in anonymised and aggregated form. Fig. 1 locates the plant at city scale. Fig. 2 shows the production sequence, the energy services and electricity share of each area, the supply and compressed-air networks and where each measure acts; it is a schematic, and no architectural drawings or building models are shown. Shares refer to the audited reference month (April 2026).

Table 1 summarises the data and their status. Monthly electricity consumption, billed demand, power factor (PF) and the average price per kWh were transcribed from 28 utility bills (January 2024–April 2026, Table A.6). Monthly production in pairs was taken from plant records for 25 of these months. During a walk-through audit in the first half of 2026 the audit team estimated the distribution of electricity use across twelve production areas for a reference month (April 2026). Significant energy uses were vulcanising (29.2%), stitching (19.8%), the compressed-air room (16.8%) and sole injection moulding (9.1%) (Fig. 4a). Tariff charges for every month from January 2024 to October 2026 were retrieved from the utility (Comisión

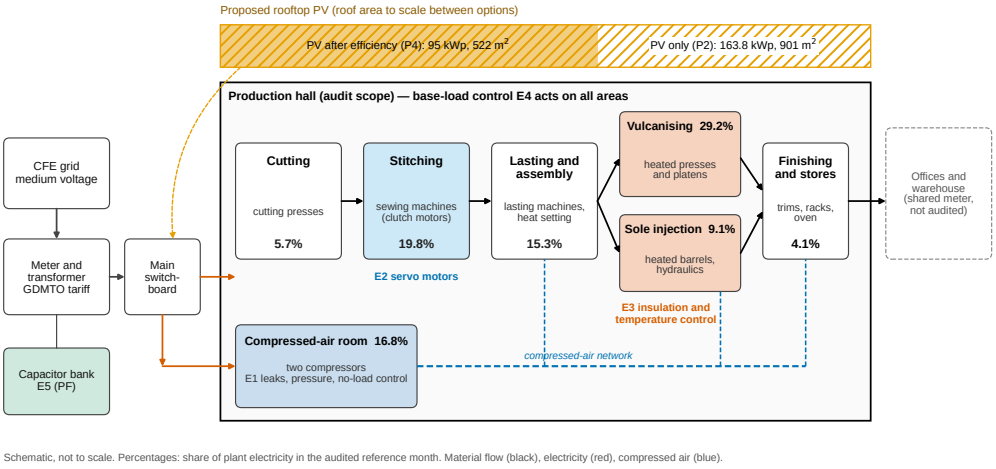


Figure 2: Plant layout and energy flows (schematic, not to scale). Percentages: share of plant electricity; E1–E5: measures.

Federal de Electricidad, 2026) (Table A.7). The emission factor, fuel prices, grid mix and solar resource were taken from official or institutional sources (Table 1). Monetary values are in 2026 Mexican pesos (MXN; kMXN and MMXN denote thousands and millions) excluding value-added tax (VAT), which firms recover. For reference, 1 USD = 17.10 MXN (Comisión Nacional de Energía, 2026).

3.2. Energy baseline and performance indicator

Following ISO 50006 (ISO, 2023), the energy baseline (EnB) relates monthly electricity use E_m to production P_m through ordinary least squares, $E_m = a + b\,P_m + \varepsilon_m$. The intercept a is interpreted as production-independent (base) load. Three specifications were estimated and compared with the regression originally reported by the audit team: all 25 months; 22 months excluding the documented 24-hour-shift months; and the original 16-month subset (April 2024–June 2025 plus September 2025). Model adequacy is reported through R^2 and the coefficient of variation of the root-mean-square error (CV-RMSE). The specification that excludes the 24-hour-shift months is used to define the base load targeted by the operational measure (E4).

Table 1: Data sources and their status.

Item	Content	Source / status
Electricity bills	kWh, billed kW, PF and average price; Jan 2024–Apr 2026 (28 months)	Utility bills, transcribed (measured)
Production	Pairs per month; Apr 2024–Apr 2026 (25 months)	Plant records (measured)
End-use split	Share of 12 areas in the reference month	Audit walk-through estimate
Tariff	GDMTO fixed, energy, distribution and capacity charges, Centro Oriente, monthly 2024–2026	Comisión Federal de Electricidad (2026); Comisión Reguladora de Energía (2025) (official)
Billed-demand rule	$\min\{D_{\max}, Q/(24\,d\,F_C)\}$, $F_C = 0.55$; PF surcharge and bonus	Comisión Reguladora de Energía (2025) (official)
Grid emission factor	0.444 tCO_2e MWh^{-1} (2024), still mandated for 2026 reporting	Secretaría de Medio Ambiente y Recursos Naturales (2025, 2026) (official)
Grid mix and gas imports	Gas-fired share of 2024 generation; US pipeline imports vs. consumption	Secretaría de Energía (2025); U.S. Energy Information Administration (2025)
Fuel prices	Natural-gas wholesale index (12-month mean); LP-gas maximum price for Pachuca	Comisión Nacional de Energía (2026); Procuraduría Federal del Consumidor (2026) (official)
Solar resource	Monthly global and diffuse irradiation, 2001–2025 climatology	NASA Langley Research Center (2026)
Measure costs	Capital costs of PV, capacitor bank and compressed-air works	Audit quotations; other costs assumed (Table B.8)

3.3. Tariff model and validation

The GDMTO bill (pre-VAT) in month m is

$$B_m = \left[c_f + c_e\, Q_m + (c_d + c_c)\, D_m^b\right](1+\phi_m), \qquad D_m^b = \min\left\{D_m^{\max},\ \frac{Q_m}{24\, d_m\, F_C}\right\}, \tag{1}$$

where c_f is the fixed charge, c_e the energy charge, c_d and c_c the distribution and capacity charges per kW, Q_m the energy consumed, d_m the days in the billing period and $F_C = 0.55$ the load factor prescribed for GDMTO (Comisión Reguladora de Energía, 2025). Because GDMTO has no peak period, the capacity charge always applies the energy-based term; the distribution charge applies the smaller of measured maximum demand and the energy-based term (Comisión Federal de Electricidad, 2026). The PF adjustment ϕ_m is a surcharge of $\frac{3}{5}(90/\mathrm{PF} - 1)$ below 90% PF and a bonus of $\frac{1}{4}(1 - 90/\mathrm{PF})$ at or above 90%, capped at 120% and 2.5% respectively (Comisión Reguladora de Energía, 2025).

The model was validated by applying month-specific official charges to the recorded consumption, billed demand and PF of all 28 months and comparing the result with the average price reported on each bill. The reported billed demand closely matches the energy-based term $Q_m/(24 d_m F_C)$ (mean ratio 1.004). We therefore model billed demand with the energy-based term in all scenarios. Where $Q_m/(24 d_m F_C)$ falls below the measured peak, as appears to be the case here, a kWh saved reduces both the energy charge and the two demand charges. The avoided cost per kWh is then

$$v = c_e + \frac{c_d + c_c}{24\, \bar{d}\, F_C}, \tag{2}$$

evaluated at the October 2026 charges and an average month length $\bar{d} = 30.4$ days. All scenarios use these charges with constant real prices. The reference year is May 2025–April 2026, the latest twelve billed months (460.8 MWh).

3.4. Options appraised

Demand-side measures. Five measures derived from the audit were modelled (Table B.8). They are applied sequentially to avoid double counting: E4 first scales all end uses, then component measures act on the remaining energy of their target areas.

- **E1 Compressed-air package**: leak detection and repair, reduction of discharge pressure and no-load control of the two compressors. The saving is 10–30% of compressed-air energy (central 20%) (U.S. Department of Energy and Compressed Air Challenge, 2016). Capital cost follows the audit estimate, plus an annual ultrasonic leak survey.

- **E2 Servo motors on sewing machines**: clutch motors in the stitching area are replaced with servo motors that save 50–75% of motor energy (central 65%) (Nemeša, 2023). Motors are assumed to use 50–85% of stitching-area energy. The number of machines (63 in the central case) follows from the motor energy, a clutch-motor running power of 0.25–0.45 kW and the scheduled operating hours. It is multiplied by an installed unit cost.

- **E3 Insulation of heated platens and barrels**: insulating jackets and improved temperature control on vulcanising presses and injection-unit barrels. The saving is 10–30% of heater energy (central 20%) (Kent, 2018). Heaters are assumed to account for 50–80% of the energy of these areas.

- **E4 Base-load (idle) control**: shutdown routines, timers and sub-metering that cut production-independent load by 5–20% (central 10%) (Mouzon et al., 2007; Kara and Li, 2011). The base load is taken as the EnB intercept.

- **E5 Power-factor correction**: the 50 kVAr automatic capacitor bank proposed in the audit, raising PF to 0.95–0.99 (central 0.97). It saves money through the PF adjustment but not energy.

Supply-side options. **S1 Rooftop PV** is the 163.8 kWp array (252 modules of 650 W) proposed in the audit, connected under the distributed-generation regime with monthly net metering. Monthly plane-of-array irradiation for a south-facing 20° tilt was computed from NASA POWER global and diffuse irradiation using the isotropic-sky monthly method of Klein (1977). Generation applies a performance ratio of 0.75–0.85 (Reich et al., 2012) and 0.5% per year degradation. Net metering reduces Q_m and therefore the energy-based billed demand. A regulatory stress case removes this demand credit, so that PV offsets only the energy charge. Monthly surplus, if any, is conservatively assigned zero value.

S2 Fuel substitution and electrification of heat is assessed parametrically, per GJ of useful heat, because the plant's thermal fuel use was not metered during the audit. Five supply routes are compared: liquefied petroleum gas (LPG) or natural-gas boilers (efficiency 0.85), electric resistance, heat pumps (COP 2.5–4.0; Arpagaus et al., 2018; Schlosser et al., 2020) and recovery of compressor heat; more than 80% of compressor input power becomes available heat (U.S. Department of Energy and Compressed Air Challenge, 2016), and we conservatively assume that 50–80% of input power (central 70%) is recovered. Prices are the LPG maximum price for Pachuca in October 2026 (Procuraduría Federal del Consumidor, 2026) and the twelve-month mean of the national natural-gas wholesale index (Comisión Nacional de Energía, 2026), doubled to approximate transport and distribution to an end user. Electricity is valued with Eq. (2). Combustion emission factors follow IPCC (2006). Because the plant's thermal fuel use is unknown, results are reported per GJ of useful heat and can be scaled by any annual fuel consumption.

3.5. Common indicators

Each option and portfolio is evaluated with the same indicators:

- *Energy*: annual grid purchases and the share avoided.
- *Emissions*: grid purchases multiplied by the official factor of 0.444 $tCO_2e\ MWh^{-1}$, held constant because the official series rose over 2022–2024 (Comisión Reguladora de Energía, 2023, 2024; Secretaría de Medio Ambiente y Recursos Naturales, 2025). A decline of 3% per year would lower lifetime avoided emissions by 24% but would not change any monetary result.
- *Cost of conserved energy*: $\mathrm{CCE} = (I \cdot \mathrm{CRF}_{r,L} + C_{\mathrm{O\&M}})/\Delta E$ (Meier et al., 1982), where $\mathrm{CRF}_{r,L}$ is the capital recovery factor at discount rate r over life L. For PV the analogous metric is the levelised cost of electricity (LCOE). Both are compared with the avoided tariff v.
- *Economics*: NPV over 20 years at a 10% real discount rate, with like-for-like reinvestment of short-lived measures, straight-line residual values and a PV inverter replacement in year 12; simple payback; and the net abatement cost, defined as the annualised negative NPV per tonne of CO_2.

- *Security and resilience*:
  - imported natural gas avoided upstream, $\Delta E_{\text{grid}} \cdot s_{\text{gas}} \cdot h \cdot s_{\text{imp}}$, with gas-fired share of generation $s_{\text{gas}} = 0.638$ (Secretaría de Energía, 2025), heat rate $h = 7.2$ GJ MWh^{-1} and import share $s_{\text{imp}} = 0.74$ (U.S. Energy Information Administration, 2025);
  - exposure to an energy-price shock, measured as the increase in the annual bill if the energy charge rose by 50%;
  - the highest energy-implied billed demand, a proxy for proximity to the 100 kW threshold above which GDMTO customers must migrate to the time-of-use GDMTH tariff (Comisión Federal de Electricidad, 2026); PV is not credited with peak reduction because cloudy days and evening operation make its contribution to the measured peak unreliable;
  - the roof area required by PV;
  - the sensitivity of value to a change in distributed-generation compensation.

3.6. Portfolios and efficiency-first pathways

Four portfolios are compared with the baseline (P0): P1, the efficiency bundle (E1–E5); P2, supply only (the 163.8 kWp array); P3, both; and P4, efficiency first, in which the efficiency bundle is implemented and PV is then sized to reach the same grid purchases as P2. To generalise P4, we construct pathways for targets of 5–75% reduction in grid purchases. The supply-only pathway sizes PV directly to the target (with PF correction in both pathways for fairness). The efficiency-first pathway adds measures in ascending order of CCE until the target is met and closes any remaining gap with PV.

3.7. Uncertainty analysis

Parameter uncertainty is propagated with 5,000 Monte Carlo draws from triangular distributions whose bounds are listed in Table B.8. These cover measure savings and costs, PV cost, performance and degradation, discount rate (6–14%) and real tariff escalation (-1 to $+2$% per year). The grid-gas parameters do not affect monetary results; their ranges are reported for the imported-gas indicator as low and high bounds. We report 5th–95th percentiles, the probability that each portfolio has positive NPV, the probability that P4 outperforms P2 and Spearman rank correlations between

inputs and the NPV difference P4−P2. One-way sensitivities are given in Appendix D. The demand-credit rule for PV is treated as a discrete regulatory scenario rather than a random variable.

3.8. Net-zero scenarios with renewables and storage

Two net-zero definitions are examined because they lead to different designs. *Annual net-zero electricity* requires on-site PV generation to equal the plant's annual consumption, with the grid used as a seasonal buffer. Under the distributed-generation rules, net-metered surpluses are credited in kWh against later bills for up to 12 months, after which unused credits are paid at the local marginal price (Comisión Reguladora de Energía, 2017). We conservatively value expired credits at zero. *Hourly self-sufficiency* measures the share of hourly consumption met on site by PV and a lithium-ion battery, a proxy for physical independence and resilience to supply interruptions. Targets of 70%, 80% and 90% are analysed.

Because interval data are not available, an hourly load profile was synthesised for each reference-year month. It consists of an always-on load of 15 kW (range 8–25 kW, an assumption) and a flat load during scheduled operating hours that closes the monthly energy balance; during the three 24-hour-shift months production is spread over Monday–Saturday. Base-load control (E4) is assumed to reduce the always-on load first. Hourly PV output uses NASA POWER 2023 hourly global and diffuse irradiance at the site (NASA Langley Research Center, 2026), transposed to the array plane with an isotropic-sky model and rescaled month by month to the 2001–2025 climatology, which gives a specific yield of 1,523 kWh kWp^{-1}, consistent with the monthly model. The battery follows a greedy self-consumption dispatch, with 85% round-trip efficiency, a 15-year life and fixed O&M of 2.5% of capital per year (National Renewable Energy Laboratory, 2024), 90% usable depth of discharge, a power rating of 0.5 kW per kWh and an installed cost of 8,500 MXN kWh^{-1} (an assumption anchored on the pack prices of about 200 USD kWh^{-1} reported by National Renewable Energy Laboratory, 2024; its influence is examined through a break-even analysis).

For each portfolio (supply only, or efficiency bundle first) we sweep PV capacity from 0.6 to 2.4 times the annual net-zero size, capped at the 0.7 MW distributed-generation limit (Cámara de Diputados, 2025), and battery capacity from 0 to 2,400 kWh, and retain the least-capital design that meets each self-sufficiency target. Each design is valued under three compensation regimes: monthly net metering with energy-based billed demand (the central

case); net metering in which billed demand follows gross imports rather than net consumption; and net billing, in which imports are billed at the tariff and exports are paid 1.0 MXN kWh^{-1}. PV degradation of 0.5% per year is applied to the PV share of the savings. Battery cost is explored through a break-even analysis; the other parameters of the net-zero module are held at their central values except the always-on load.

The five key designs were cross-validated in EnergyPlus 25.1. An EPW file was built from the same rescaled NASA POWER hourly data, converted from local solar to local standard time; PV was modelled with `Generator:PVWatts` (version 5, 14% system losses, roof-mounted, 20° tilt, south) and a PVWatts inverter (DC/AC ratio 1.1, 96% efficiency); storage with `ElectricLoadCenter:Storage:Simple` using the same capacity, power and efficiencies, AC-coupled and dispatched to track facility demand and store on-site surplus. The same hourly load was imposed through a schedule file, and EnergyPlus hourly purchased and surplus electricity were passed through the same net-metering bank and tariff.

3.9. Reproducibility, data and code

The model is written in Python (NumPy, pandas, Matplotlib) and organised as a small package with scripts that regenerate every number, table and figure in this paper from the input data. All numbers quoted in the text are inserted automatically from the results files. The public repository contains the model code, the official tariff, price, emission-factor and climate series with their sources and access dates, all derived results files, the figures, the Monte Carlo seed and a README with run instructions. The plant's billing, production and end-use records are confidential and are not included; the repository documents their format so that the scripts can be run with them, or with another plant's data. Code is released under the MIT licence and public data, results and figures under CC BY 4.0. The repository is archived on Zenodo (`https://doi.org/10.5281/zenodo.23419738`) and mirrored on GitHub (`https://github.com/kerdan85/efficiency-first-sme`).

To make the method usable beyond this plant, the same repository contains `ee1st`, a general-purpose implementation installable as a Python package from its `ee1st/` folder (Alberú Argüeso et al., 2026). A plant is described in a single case file that lists its bills, tariff, end-use split, candidate measures and PV and storage options; every numeric input can be given as a central value or as a (central, low, high) range with a data-status label (for example measured, official, literature, quote or assumption). From this file the toolkit

produces the core indicators, portfolios, pathways, Monte Carlo statistics and net-zero designs described in Sections 3–4, together with a short report; the EnergyPlus cross-check is not part of it. Three tariff structures are supported: Q-based demand charges as in GDMTO, energy plus measured-peak demand charges as common in Europe and North America, and flat energy prices. Measures are declared as base-load, end-use or power-factor measures, and a production series is optional; without it the base load is supplied directly. The footwear case is included as a configuration file with its public inputs; when the confidential plant records are supplied, an automated test confirms that the toolkit reproduces the deterministic results of this paper exactly and the Monte Carlo statistics within sampling error. An explicitly synthetic example, which runs without any confidential data, shows the inputs for a plant under an energy-plus-demand tariff without production records.

4. Results

4.1. Energy use, baseline and tariff

Monthly consumption ranged from 19,040 to 64,160 kWh (mean 35,523 kWh), and billed demand reached 157 kW (Fig. 3a). Power factor averaged 87.1% and fell below the 90% threshold in 25 of 28 months, incurring net PF adjustments of about 13.7 kMXN in the reference year at 2026 charges. Production explains only part of the variability (Fig. 3b). Over all 25 months the regression gives $E = 24{,}349 + 0.82\,P$ ($R^2 = 0.29$, CV-RMSE 26.7%). Excluding the 24-hour-shift months yields $E = 25{,}341 + 0.54\,P$ ($R^2 = 0.38$, CV-RMSE 16.0%). The regression in the audit report ($E = 20{,}223 + 0.90\,P$, $R^2 = 0.72$) is reproduced only with the non-contiguous 16-month subset and with one month transcribed as 34,200 instead of 32,400 kWh. With the corrected value the subset gives $R^2 = 0.70$. Whatever the specification, the intercept is large: base load accounts for about 66% of the reference-year mean consumption. Much of the plant's electricity is therefore not proportional to output, a typical signature of idle equipment, compressed-air leakage and heaters kept at temperature (Kara and Li, 2011; Gutowski et al., 2009).

The tariff model reproduces the reported average price of each bill with a mean bias of 2.1% and a mean absolute error of 2.2% (maximum 4.7%; Fig. 3c). The small positive bias is consistent with minor charges not itemised in the transcribed data. Two consequences follow. First, the plant's billed demand is set by its monthly energy rather than its instantaneous peak. Under October 2026 charges, an avoided kWh is therefore worth $v = 2.64$ MXN

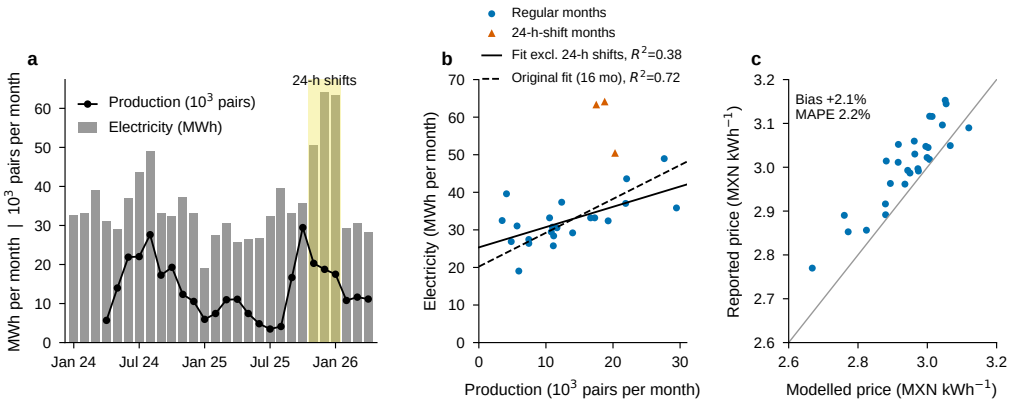


Figure 3: Energy data, baseline and tariff validation: (a) monthly electricity and production (shading: 24-hour shifts); (b) baseline regressions; (c) reported versus modelled price.

(0.155 USD), of which only 56% is the energy charge itself. Second, billed demand exceeded 100 kW in 5 months, including three consecutive months during the 24-hour shifts. The load factors printed on the bills, if defined on measured maximum demand, imply measured peaks of 121–190 kW, above the 100 kW GDMTO limit in 28 of 28 months. The plant therefore appears to qualify for reclassification to the time-of-use GDMTH tariff, a regulatory exposure discussed in Section 5. The reference-year bill is 1.23 MMXN.

4.2. Cost of conserved energy of demand-side measures

Table 2 and Fig. 4b rank the measures. Base-load control (E4) is by far the cheapest resource, at 0.21 MXN kWh^{-1} for 30.4 MWh per year. It is followed by insulation of heated platens and barrels (E3, 0.82), servo motors (E2, 0.94, the largest single saving at 41.5 MWh) and the compressed-air package (E1, 1.04). At central values all four lie well below the avoided tariff ($v = 2.64$) and also below the energy charge alone (1.469). Their 5th–95th percentile ranges, from 0.11–0.37 for E4 to 0.78–1.74 for E1, never reach the avoided tariff, and all measures remained below it simultaneously in 99.98% of Monte Carlo draws. Even if only the energy charge were avoided, E1 would still be below it in 85% of draws and the other measures in at least 95%. The measures are of the same order as the PV LCOE (0.92 MXN kWh^{-1}, range 0.74–1.20). E4 was cheaper than PV in 100% of draws and E3 in 65%, whereas E2 (36%) and E1 (25%) were usually slightly more expensive per

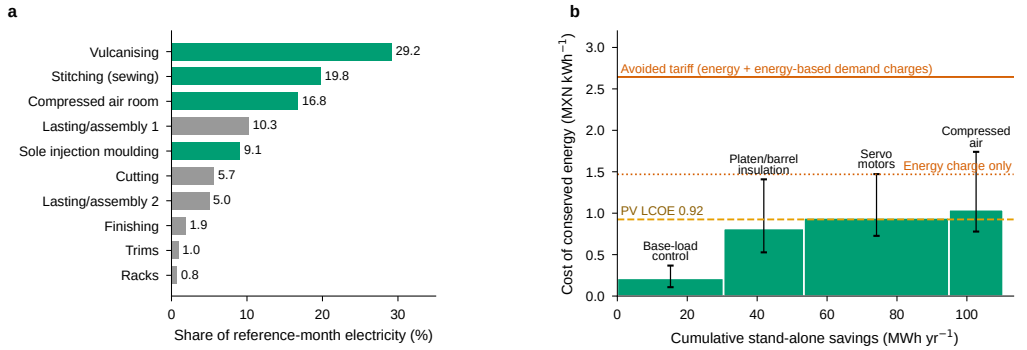


Figure 4: End uses and conservation supply curve: (a) electricity share by area; (b) CCE of each measure (whiskers: 5th–95th percentiles).

kWh than PV. The distinctive value of the latter two lies less in their unit cost than in their small capital requirement and their contribution to lower demand (Section 4.5). Power-factor correction saves no energy but removes the PF surcharge and earns the bonus, returning an NPV of 0.21 MMXN on 85 kMXN.

Individually, every measure has a payback of 0.5–2.4 years (Table 2). Combined (P1), they reduce grid purchases by 105.1 MWh per year (23%), avoid 47 tCO_2 per year and require 0.52 MMXN (30 kUSD) for an NPV of 1.84 MMXN.

Table 2: Stand-alone appraisal of each option (CCE: for PV, LCOE; 5th–95th percentiles in brackets).

Option	Capex (kMXN)	Electricity saved (MWh yr^{-1})	CCE / LCOE (MXN kWh^{-1})	Bill saving, yr 1 (kMXN yr^{-1})	NPV (MMXN)	Payback (yr)	CO_2 avoided (t yr^{-1})
E4 Base-load (idle) control	40	30.4	0.21 [0.11–0.37]	82	0.64	0.5	13.5
E3 Insulation of heated platens/barrels	100	23.0	0.82 [0.53–1.41]	61	0.36	1.6	10.2
E2 Servo motors on sewing machines	241	41.5	0.94 [0.73–1.47]	111	0.61	2.2	18.4
E1 Compressed-air package	50	15.5	1.04 [0.78–1.74]	41	0.21	1.5	6.9
E5 Power-factor correction (50 kVAr)	85	0.0	–	36	0.21	2.4	0.0
S1 Rooftop PV (163.8 kWp)	1,802	250.0	0.92 [0.74–1.20]	670	3.57	2.8	111.0

4.3. Portfolio comparison

Table 3 and Fig. 5 compare the portfolios. Supply-only PV (P2) generates 250 MWh per year with a specific yield of 1,526 kWh kWp^{-1}. It cuts grid purchases by 54% and has a larger absolute NPV (3.57 MMXN) than the

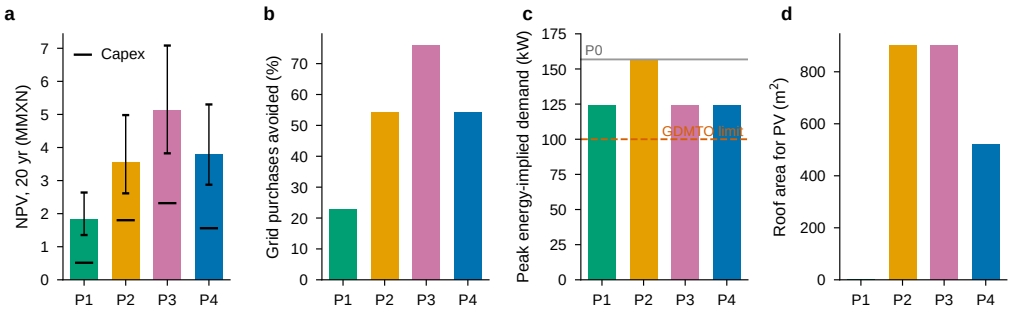


Figure 5: Portfolio indicators: (a) NPV and capital cost; (b) cut in grid purchases; (c) peak billed demand; (d) PV roof area.

efficiency bundle (1.84 MMXN), because it is a larger investment. Per peso invested, however, efficiency returns more: NPV/capex is 16.0 for E4 and between 2.5 and 4.3 for E1–E3, against 2.0 for PV, and P1 beats P2 on this ratio in 99% of Monte Carlo draws.

The efficiency-first portfolio (P4) reaches the same grid purchases as P2 (211 MWh per year, within 0.07 MWh) with 94.9 kWp of PV instead of 163.8 kWp. It requires 13% less capital (1.56 versus 1.80 MMXN) and 42% less roof (522 versus 901 m^2), and achieves a higher NPV (3.80 MMXN, +0.24 MMXN). Emissions fall by 111 t per year in both cases (54% of the baseline 205 t). P4 also lowers the energy-implied peak from 157 to 124 kW, which PV alone cannot do. Installing the efficiency bundle together with the full 163.8 kWp array (P3) cuts grid purchases by 76% and emissions by 155 t per year, with the largest NPV (5.11 MMXN) but also the largest capital requirement (2.32 MMXN). Net abatement costs are negative for every portfolio, between −4,632 and −3,773 MXN tCO$_2^{-1}$.

Table 3: Portfolio comparison on common indicators.

Portfolio	Capex (MMXN)	NPV (MMXN)	Grid purchases (MWh yr^{-1})	Reduction (%)	Residual CO$_2$ (t yr^{-1})	Peak demand (kW)	Shock exposure (kMXN yr^{-1})	Imported gas (GJ yr^{-1})	PV roof (m^2)
P0 Baseline	0.00	0.00	461	0	205	157	342	0	0
P1 Efficiency bundle (E1–E5)	0.52	1.84	356	23	158	124	257	357	0
P2 Supply only: PV 163.8 kWp	1.80	3.57	211	54	94	157	156	850	901
P3 Efficiency bundle + PV 163.8 kWp	2.32	5.11	111	76	49	124	80	1,188	901
P4 Efficiency first + PV 94.9 kWp	1.56	3.80	211	54	94	124	152	850	522

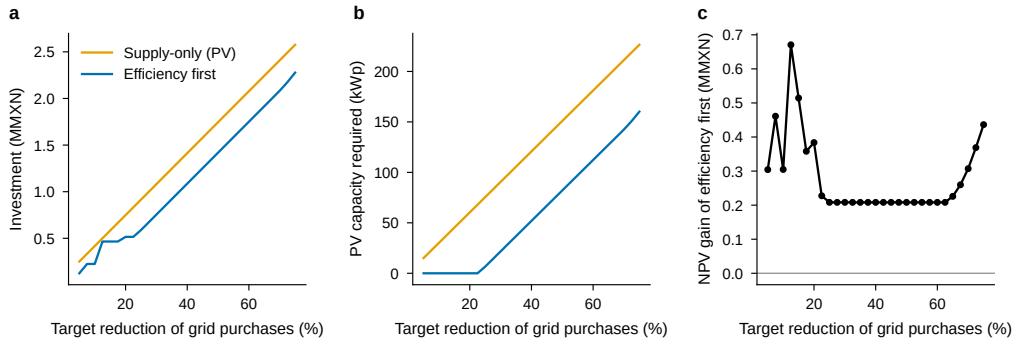


Figure 6: Supply-only versus efficiency-first pathways: (a) capital; (b) PV capacity; (c) NPV gain of efficiency first.

4.4. Efficiency-first pathways

Fig. 6 generalises the comparison across reduction targets. Up to the 23% reduction delivered by the efficiency bundle, efficiency measures alone meet the target and no PV is needed. Beyond that point PV is still required, but the array is always smaller than in the supply-only pathway. At a 50% target, for instance, the efficiency-first pathway needs 82 kWp instead of 151 kWp, 1.42 instead of 1.75 MMXN of capital (19% less) and 451 instead of 830 m^2 of roof, with an NPV gain of 0.21 MMXN. The NPV gain is positive for every target. It is largest where efficiency alone meets the target and remains roughly constant at intermediate targets, where both pathways add PV at the same marginal value. It rises again above about 65%, where the supply-only array starts to produce monthly surpluses that net metering does not remunerate in this model. The steps at low targets arise because measures are added as discrete units in merit order and can exceed the target.

4.5. Energy security and resilience

The security indicators favour efficiency in ways that the averaged economics do not reveal (Fig. 7).

- *Regulatory stress.* If distributed-generation output no longer reduced billed demand, for example under a move from net metering to net billing, the NPV of the PV array would fall by 69% to 1.11 MMXN. The NPV of the efficiency bundle would be unchanged (1.84 MMXN), because its savings reduce consumption behind the meter.

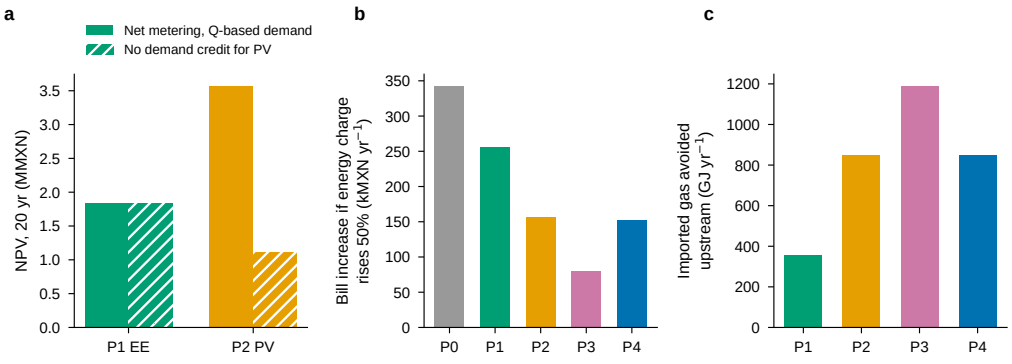


Figure 7: Resilience indicators: (a) NPV with and without PV demand credit; (b) bill increase if the energy charge rises 50%; (c) imported gas avoided.

- *Energy-price shock.* A 50% increase in the energy charge would raise the baseline bill by 342 kMXN per year. The efficiency bundle cuts this exposure by 25%, and P4 cuts it by 56%, at least as much as P2 but with less capital.

- *Imported gas.* Upstream, the grid electricity avoided by P1 corresponds to about 357 GJ per year of imported natural gas (287–474 GJ across the parameter ranges), and that avoided by P2 and P4 to 850 GJ (683–1,127 GJ).

- *Tariff-class threshold.* Efficiency reduces the highest energy-implied billed demand by 32 kW. Neither this indicator in the surge months nor the measured peaks implied by the bills fall below the 100 kW GDMTO limit, so peak management (scheduling of compressors, presses and heaters) would be needed if the plant is reclassified.

4.6. Fuel substitution and electrification of heat

Per GJ of useful heat, an LPG boiler costs 424 MXN and emits 74 $kgCO_2$. A natural-gas boiler costs 126 MXN and emits 66 $kgCO_2$ (Fig. 8). Electric resistance heating at the avoided tariff costs 734 MXN GJ^{-1} and emits 123 $kgCO_2$ GJ^{-1}, more than either fuel on both counts. A heat pump with COP 3 costs 245 MXN and emits 41 $kgCO_2$ per GJ. It beats LPG on cost for any COP above 1.7 and on emissions above 1.7. Against pipeline natural gas,

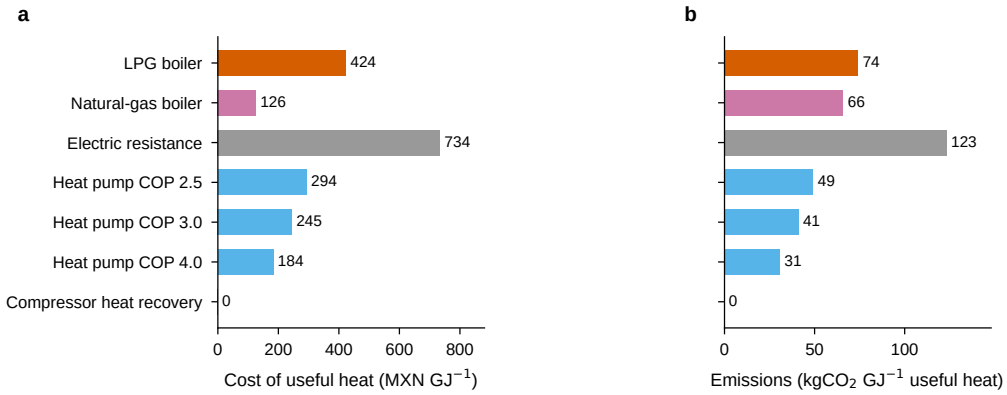


Figure 8: Heat supply routes: (a) cost and (b) emissions per GJ of useful heat.

however, a heat pump breaks even on emissions at COP 1.9 but on cost only at COP 5.8, because wholesale gas is cheap relative to GDMTO electricity. Electrification is therefore a strong option where the plant burns LPG and a weak one where pipeline gas is available, unless electricity prices or gas prices change.

Efficiency interacts with electrification in two ways. First, it creates headroom: the 105.1 MWh saved each year by P1 could supply about 1,135 GJ of useful heat through a COP-3 heat pump without increasing grid purchases above the baseline. Second, it reduces the heat available for recovery. Compressor heat that could be recovered falls from about 195 to 146 GJ per year once E1 and E4 are implemented. Recovery projects should therefore be sized after the compressed-air system has been optimised, not before.

4.7. Uncertainty

All portfolios had positive NPV in every Monte Carlo draw (Fig. 9a). The 5th–95th percentile ranges were 1.35–2.64 MMXN for P1, 2.61–4.98 for P2 and 2.88–5.30 for P4. P4 outperformed P2 in 86% of draws. The advantage of efficiency first is driven mainly by the achievable base-load saving ($\rho = 0.67$), followed by the insulation saving ($\rho = 0.34$) and, at similar magnitude, the servo saving ($\rho = 0.27$) and PV cost ($\rho = 0.27$). It shrinks when PV performs better than expected ($\rho = -0.21$; Fig. 9b). One-way sensitivities (Appendix D) confirm that no single parameter within its range reverses the sign of any portfolio NPV.

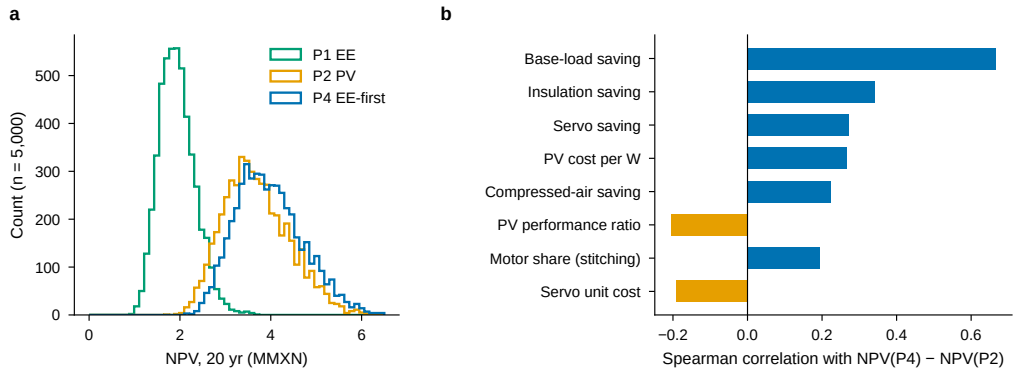


Figure 9: Uncertainty (5,000 draws): (a) NPV distributions; (b) main drivers of the NPV difference P4 − P2.

4.8. Pathways to net-zero electricity

Annual net-zero electricity is technically and economically within reach (Fig. 10). Without efficiency, the plant needs 303 kWp of PV on 1,665 m^2 of roof (3.41 MMXN). With the efficiency bundle first it needs 234 kWp on 1,285 m^2 (3.09 MMXN, including the measures), that is 69 kWp and 380 m^2 less, and its NPV is higher (6.65 versus 6.44 MMXN). Under net metering with a 12-month credit bank the annual bill falls to the fixed charge of about 4.0 kMXN, and no credits expire, because the seasonal surplus of the dry months is used during the high-consumption winter months (Fig. 10c). Both designs stay well below the 0.7 MW distributed-generation limit. Roof space is the binding physical constraint. The usable roof area is not documented; the audit's 252-module layout implies at least about 901 m^2. On that area the full efficiency bundle plus the 163.8 kWp array (P3) cuts grid purchases by 76%, against 54% for the array alone, whereas annual net zero would need about 380 m^2 more roof without efficiency than with it. The less roof a plant has, the more efficiency first determines how close it can get to net zero.

Annual net zero is not physical independence. Without storage only 57% of hourly consumption is met on site, and gross grid imports remain 152 MWh per year in the efficiency-first design (67 tCO_2 on a location basis) and 199 MWh in the supply-only design. Above about 57% self-sufficiency storage is needed, from about 57–70% the least-cost designs also oversize the array, and the cost rises steeply (Fig. 10a). The least-capital efficiency-first

design for 80% self-sufficiency combines 350 kWp with 300 kWh (6.92 MMXN, NPV 1.80 MMXN); the supply-only equivalent needs 424 kWp and 400 kWh, 1.23 MMXN more, and has an NPV of only 0.39 MMXN, which turns negative if billed demand follows gross imports (−0.47 MMXN). At 90% the efficiency-first design needs 467 kWp and 800 kWh (12.46 MMXN) and its NPV turns negative (−5.35 MMXN). The highest self-sufficiency within the range explored (PV up to 2.4 times the annual net-zero array or 0.7 MW, and up to 2,400 kWh of storage) is about 98%, with 561 kWp and 2,400 kWh. Efficiency first lowers the capital required at every self-sufficiency level. The advantage comes mainly from lower total consumption, which shrinks the array: a supply-only design given the lower night load of the efficiency-first plant still needs 8.15 MMXN for 80% self-sufficiency.

Storage does not pay under the current rules. Under net metering the grid already acts as a free, lossless seasonal store, so adding 200 kWh to the efficiency-first net-zero design raises self-sufficiency by 14 percentage points but changes NPV by −2.49 MMXN. The loss shrinks if billed demand follows gross imports (−1.93 MMXN) or under net billing (−1.72 MMXN), but remains negative at current battery costs (Fig. 10b). Storage would break even only with battery costs 75% lower under net billing or 85% lower under import-based demand charges, and never under net metering, where round-trip losses add to net imports. The value of storage for this plant is therefore resilience: the 80% design holds about 27 hours of the always-on load. These results depend on the synthetic profile. Self-sufficiency of the annual net-zero designs ranges from 51% to 60% across the assumed always-on loads.

The EnergyPlus runs verify the PV yield and the battery dispatch of the Python model; they use the same synthetic load, weather, credit bank and tariff, so they do not test those inputs (Table 4). For the designs with storage, self-sufficiency differs by at most 0.5 percentage points and grid imports by at most 5.5%; for the annual net-zero designs without storage the difference is at most 0.2 points. EnergyPlus DC output is within 0.2% of the Python AC output, and its AC yield (1,456 kWh kWp^{-1}) is −4.4% lower mainly because the PVWatts inverter efficiency (96%) is applied on top of 14% system losses, whereas the Python model uses a single performance ratio of 0.80. Sized with the EnergyPlus yield, annual net zero requires 244 kWp with efficiency first and 316 kWp without it (a gap of 72 kWp), and the NPVs become 6.52 and 6.27 MMXN, so the ranking is unchanged.

Table 4: Net-zero designs: Python model versus EnergyPlus 25.1 (annual values).

Design	PV (kWp)	Battery (kWh)	PV output (MWh)	Grid imports (MWh)	Exports (MWh)	Self-sufficiency (%)
NZ-S annual net zero, supply only	302.7	0	461 / 441	199 / 200	199 / 180	56.8 / 56.6
NZ-E annual net zero, efficiency first	233.6	0	356 / 340	152 / 153	152 / 137	57.3 / 57.1
80% self-sufficiency, supply only	423.7	400	645 / 617	91 / 89	260 / 229	80.3 / 80.7
80% self-sufficiency, efficiency first	350.4	300	534 / 510	71 / 70	238 / 213	80.0 / 80.4
90% self-sufficiency, efficiency first	467.2	800	711 / 680	35 / 33	374 / 341	90.2 / 90.7
NZ-S re-sized with EnergyPlus yield	316.5	0	– / 461	– / 197	– / 197	– / 57.2
NZ-E re-sized with EnergyPlus yield	244.3	0	– / 356	– / 150	– / 150	– / 57.7

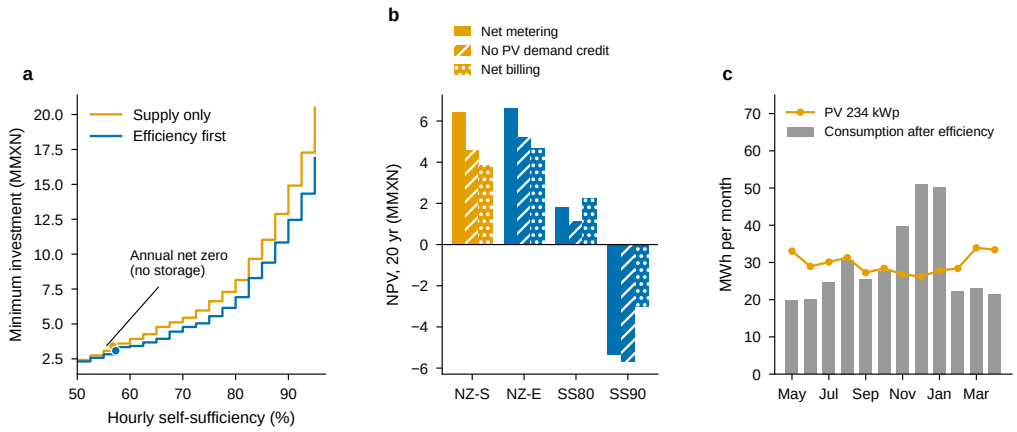


Figure 10: Net zero with PV and storage: (a) least capital by self-sufficiency target; (b) NPV under three compensation regimes; (c) monthly balance of the efficiency-first net-zero design.

5. Discussion

5.1. What efficiency first means at plant scale

The results do not support a simple "efficiency instead of solar" narrative. Rooftop PV is a profitable investment for this plant, and the largest NPV is obtained by doing both. What EE1st changes is the order and size of the supply investment. Treating demand-side measures as a resource that competes with PV, as the principle requires (Mandel et al., 2022; European Commission, 2021), reveals that efficiency delivers the first 23% of the reduction in grid purchases at low cost. It also shrinks the PV array needed for any higher target, by 69 kWp in this case. That frees capital, roof area and grid-connection capacity for later steps such as heat electrification. The same logic underpins system-level EE1st analyses (Mandel et al., 2023; Pató and Mandel, 2022). Here it appears in the investment decision of a single SME.

The low cost of the operational measure (E4) deserves emphasis. Its value rests on the large production-independent load revealed by a simple baseline regression. Such loads are common in discrete manufacturing (Kara and Li, 2011; Mouzon et al., 2007) but invisible in averaged indicators such as kWh per pair. Capturing these savings requires routines, sub-metering and accountability rather than capital, which is why energy management practices and ISO 50001 matter for SMEs (Thollander et al., 2025; McKane et al., 2017). Efficiency measures can also bring non-energy benefits, such as lower noise from compressed-air systems (Nehler, 2018). Insulating heated equipment would reduce heat gains in a plant where workers reported high indoor temperatures. These benefits are not monetised here, so the estimates for efficiency are conservative (Worrell et al., 2003; Nehler and Rasmussen, 2016).

5.2. Tariff design as a hidden driver

The tariff analysis explains why both efficiency and PV look attractive. Under GDMTO the capacity charge, and in practice also the distribution charge, is computed from monthly energy divided by a fixed load factor (Comisión Reguladora de Energía, 2025). This turns demand charges into a volumetric charge and raises the value of an avoided kWh to 2.64 MXN, 56% of which is the energy charge. For efficiency the design is benign, since saved kWh are worth the same whatever their timing. For PV it creates a regulatory dependency, because most of its value rests on net-metered generation reducing the billed demand. The stress case shows that a change in that rule would remove most of the PV's NPV while leaving efficiency untouched. An EE1st appraisal that uses only an average price would miss this asymmetry. The finding echoes the importance of incentive design stressed by Pató and Mandel (2022), seen here from the customer's side of the meter. The bills also suggest that the plant's measured peaks exceed the 100 kW GDMTO limit. Under the time-of-use GDMTH tariff, capacity is billed on demand coincident with the evening peak period and energy is priced by period (Comisión Reguladora de Energía, 2025). Base-load reductions that persist into the peak period would then gain value, whereas daytime PV output would contribute little to peak-period demand. Reclassification would therefore be likely to widen the advantage of efficiency first, although this was not simulated.

5.3. Energy security and the phase-out of fossil fuels

Because two-thirds of Mexican grid electricity is gas-fired and roughly three-quarters of the gas is imported (Secretaría de Energía, 2025; U.S. Energy Information Administration, 2025), every kWh saved or self-generated by an SME reduces the country's exposure to an external fuel supply. At the plant level, efficiency and PV reduce exposure to an energy-charge shock by similar amounts for the same reduction in grid purchases. Efficiency, however, does so without regulatory risk, without roof constraints and while lowering demand. For the fossil-fuel phase-out at the heart of this special issue, the thermal analysis adds a nuance. Electrification of heat is economically compelling against LPG but not yet against pipeline gas. Efficiency first makes room for electrification by freeing grid capacity and lowering the plant's peak. The official emission factor has not declined in recent years. Behind-the-meter efficiency is therefore the decarbonisation lever least dependent on progress in the grid.

5.4. Can the plant reach net zero?

For electricity, yes, on an annual basis and at a profit under current rules: an efficiency-first plant with about 234 kWp of rooftop PV generates as much as it consumes. Two qualifications matter for how such a claim is made. First, annual netting is not equivalent to zero emissions in a location-based inventory: about 67 tCO_2 per year of grid imports remain, mostly at night and in the winter shift months, even though exports displace a similar amount of grid generation during the day. Second, physical self-sufficiency above about 57% requires batteries whose cost is not recovered under net metering, net billing or energy-based demand charges. A net-zero claim for this plant should therefore be framed as annual net-zero electricity, with gross imports and location-based emissions reported alongside (ISO, 2018). Storage becomes attractive only if battery costs fall by at least three-quarters, if compensation rules penalise exports more than in the cases tested, or if the firm values resilience to outages explicitly. Efficiency first reduces the cost of every one of these steps, chiefly by shrinking the consumption that PV and storage must cover, and with it the array and the roof. Full net zero also requires the thermal fuels discussed in Section 4.6 to be electrified, which the electricity headroom created by efficiency makes easier.

5.5. Implications for practice and policy

For the plant, the results suggest a staged plan: (i) implement base-load control, PF correction and compressed-air maintenance immediately, as they are low-cost and verifiable; (ii) retrofit sewing motors and insulate heated equipment; (iii) size PV only after the savings have been measured; (iv) investigate thermal fuel use and heat recovery. The M&V plan prepared during the audit already specifies IPMVP options for each measure (Efficiency Valuation Organization, 2022) and is summarised in Appendix E. For policy, three implications follow. First, distributed-generation programmes and green-credit lines aimed at SMEs could require an efficiency screening, such as a simple baseline regression and an audit, before PV sizing, in the spirit of the EU's EE1st guidance (European Commission, 2024). Second, tariff reforms that move GDMTO demand charges towards measured peaks would strengthen incentives for demand management but erode the value of PV. They should therefore be signalled in advance. Third, the large share of production-independent load points to the value of energy-management networks and capacity building for SMEs (Jalo et al., 2021; Johansson et al., 2019; Herce et al., 2024).

5.6. Transferability to other SMEs

The analysis rests on data that most SMEs already hold or can obtain in a walk-through audit (Table 5). Twelve months of bills and an end-use estimate are enough to rank measures against PV, to compare the supply-only and efficiency-first pathways and to compute the security indicators. A production series adds the baseline regression that identified the base load, the cheapest resource in this case. Hourly weather adds the net-zero analysis, and interval metering would replace the synthetic load profile, the weakest element of that analysis. The toolkit lists each input with its data status and its rank correlation with the NPV advantage of the efficiency-first portfolio across the Monte Carlo draws, so that a firm can see where the next measurement is worth making. Three results are likely to transfer: base-load control is cheap wherever production-independent load is large; efficiency first shrinks the PV array whenever the PV surplus is not remunerated; and storage is hard to justify under net metering. Others depend on the tariff. In particular, the regulatory risk to PV found here arises from Q-based demand charges; under a measured-peak tariff the toolkit assumes, conservatively, that PV earns no demand credit, whereas efficiency lowers the peak in proportion to its effect on average power, so efficiency would gain relative value.

Table 5: Data tiers for applying the method to other SMEs.

Data	Analyses enabled
12 monthly bills, tariff, end-use estimate, monthly climate	Tariff check, measure CCE and NPV, merit order, portfolios P1–P4, pathways, CO_2 and security indicators, Monte Carlo
+ monthly production	ISO 50006 baseline and base-load share
+ hourly weather year	Hourly PV, battery dispatch, net-zero and self-sufficiency frontier under three compensation regimes
+ interval metering	Measured always-on load in place of the assumed value (load shapes and time-of-use tariffs are not yet implemented)
+ fuel records	Scaling the per-GJ heat-electrification results to the plant

5.7. Limitations

This is an ex-ante evaluation study, and its limits should be read alongside its findings.

- *Data transcription.* Billing data were transcribed from utility bills rather than obtained from interval metering.
- *End-use split.* The split is an audit estimate for one month and is assumed constant across months.
- *Measure savings and costs.* Savings of E1–E4 and the costs of E2–E4 come from literature and market ranges rather than plant-specific measurements and quotations. The Monte Carlo analysis shows that the conclusions hold across these ranges, but the absolute savings must be verified through the M&V plan.
- *PV and load models.* The main PV model is monthly. The net-zero analysis uses hourly PV for one weather year rescaled to climatology and a synthetic hourly load built from an assumed always-on load and scheduled operating hours; shading and actual load shapes are not resolved, and inverter clipping is represented only in the EnergyPlus runs. Both models convert the NASA POWER hourly time stamps (UTC-7 at this longitude) to local clock time; the EnergyPlus weather file assumes constant relative humidity (60%) and wind speed (2 m s^{-1}), which affect module temperature only slightly. Storage sizing in particular should be confirmed with interval data.

- *Demand ceiling.* Measured maximum demand is only implied by the load factors printed on the bills. The model assumes that the energy-based demand term remains binding, which is consistent with the billed demand.

- *Tariff migration.* Migration to GDMTH, which would apply time-of-use energy prices and capacity charges on peak-period demand, is discussed but not simulated because no interval data were available; the bills identify the plant as a GDMTO customer, which is the tariff modelled.

- *Thermal fuel use.* Because fuel consumption was not metered, the fuel-substitution analysis is parametric.

- *Energy-security indicators.* Imported gas embodied in grid electricity is estimated with average shares and an assumed heat rate rather than with marginal dispatch.

None of these simplifications favours efficiency systematically. Several are conservative, notably the exclusion of non-energy benefits and the zero value assigned to PV surplus.

6. Conclusions

This study applied the energy efficiency first principle to the investment choices of a Mexican footwear SME, comparing demand-side measures with rooftop PV and heat electrification on common indicators of energy, emissions, cost and security.

The main conclusions are:

- *Validated tariff model.* A model built from official GDMTO charges reproduces the plant's bills with a mean absolute error of 2.2%. It shows that demand charges are effectively volumetric, so each avoided kWh is worth 2.64 MXN.

- *Cost of demand-side measures.* Base-load control, insulation of heated equipment, servo motors and a compressed-air package together save 23% of grid electricity. Their central costs of 0.21–1.04 MXN kWh^{-1} are below even the energy charge alone, and their paybacks are at most 2.2 years.

- *Sequencing and PV sizing.* PV is profitable, but sequencing matters. Implementing efficiency first achieves the same 54% reduction in grid purchases with 42% less PV capacity, 13% less capital and a higher NPV in 86% of uncertainty draws.

- *Security and resilience.* Efficiency lowers the plant's peak, preserves its value if distributed-generation compensation changes and reduces exposure to imported gas and to energy-price shocks.

- *Heat electrification.* Heat pumps outcompete LPG boilers above a COP of 1.7, but not yet pipeline gas at current wholesale prices.

- *Net zero.* Annual net-zero electricity is achievable and profitable with 234–244 kWp of PV after efficiency (Python and EnergyPlus sizing; 69–72 kWp less than without it). Hourly self-sufficiency of 80% needs 300 kWh of storage and 90% turns the NPV negative; under current rules storage is a resilience investment, not a cost saving.

For SMEs in emerging economies, EE1st is best understood as a sequencing rule: measure, cut the waste, then size the supply. The open `ee1st` toolkit released with this paper allows other plants to repeat the assessment from twelve monthly bills and an end-use estimate, adding detail as production, weather or interval data become available. Future work should verify the savings ex post, add interval data to resolve hourly PV self-consumption and peak demand, and quantify thermal fuel use so that the phase-out of fossil fuels on site can be assessed directly.

CRediT authorship contribution statement

Yeniba Alberú Argüeso: Investigation, Data curation, Formal analysis, Writing – review & editing. **Ivan Aguado Guerra:** Investigation, Data curation, Methodology, Writing – review & editing. **Francisco Acosta Sosa:** Investigation, Formal analysis, Writing – review & editing. **Samuel Antonio Rosas-Meléndez:** Validation, Visualization, Writing – review & editing. **Iván García Kerdan:** Conceptualization, Methodology, Software, Validation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

Iván García Kerdan is a Guest Editor of the special issue "Energy Efficiency First for Net Zero: Enhancing Energy Security, Resilience, and Decarbonisation through the Phase-Out of Fossil Fuels" and a member of the Editorial Board of *Energy Efficiency First*. In accordance with Elsevier policy, the Editor-in-Chief was informed before submission, this is the only manuscript he has submitted to the special issue, and the manuscript was handled by another editor; he had no access to its peer-review process. The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude 5.5 (Anthropic) to a minimal extent, for language editing and checking of code; the authors reviewed and edited all content and take full responsibility for the publication.

Data availability

The plant's billing, production and audit records are confidential and cannot be shared, because the company has not authorised their release. The model code, the public input data (tariffs, emission factors, fuel prices and climate), all derived results and the scripts that produce the tables and figures are openly available on Zenodo (`https://doi.org/10.5281/zenodo.23419738`) under MIT (code) and CC BY 4.0 (data and results) licences. The same deposit contains the generalised toolkit `ee1st` (folder `ee1st/`), with the footwear plant and a synthetic plant as examples. See Section 3.9.

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Appendix A. Billing, production and tariff data

The monthly records were transcribed from utility bills and plant production records; they are confidential and only their summary is reproduced (Table A.6). Table A.7 lists the official tariff charges for Pachuca de Soto.

Table A.6: Billing and production records, January 2024–April 2026 (summary only; monthly data confidential).

Variable	Months	Minimum	Mean	Maximum
Electricity (kWh per month)	28	19,040	35,523	64,160
Billed demand (kW)	28	46	89	157
Power factor (%)	28	84.4	87.1	90.6
Average price (MXN kWh^{-1}, pre-VAT)	28	2.770	3.005	3.153
Production (pairs per month)	25	3,473	13,692	29,460

Table A.7: Official GDMTO charges, Centro Oriente division, pre-VAT (Comisión Federal de Electricidad, 2026).

Month	Fixed (MXN month^{-1})	Energy (MXN kWh^{-1})	Distribution (MXN kW^{-1})	Capacity (MXN kW^{-1})
2024-01	360.01	1.544	155.26	342.36
2024-02	360.01	1.552	155.26	344.34
2024-03	360.01	1.569	155.26	348.69
2024-04	360.01	1.569	155.26	348.70
2024-05	360.01	1.569	155.26	348.70
2024-06	360.01	1.569	155.26	348.70
2024-07	360.01	1.608	155.26	358.56
2024-08	360.01	1.621	155.26	361.67
2024-09	360.01	1.621	155.26	361.69
2024-10	360.01	1.621	155.26	361.70
2024-11	360.01	1.621	155.26	361.69
2024-12	360.01	1.597	155.26	355.67
2025-01	308.10	1.617	155.81	359.79
2025-02	308.10	1.620	155.81	360.36
2025-03	308.10	1.620	155.81	360.51
2025-04	308.10	1.620	155.81	360.51
2025-05	308.10	1.620	155.81	360.51
2025-06	308.10	1.654	155.81	369.07
2025-07	308.10	1.676	155.81	374.61
2025-08	308.10	1.678	155.81	375.13
2025-09	308.10	1.668	155.81	371.70
2025-10	308.10	1.668	155.81	371.70
2025-11	308.10	1.624	155.81	360.54
2025-12	308.10	1.553	155.81	342.52
2026-01	335.36	1.519	149.44	334.81
2026-02	335.36	1.520	149.44	335.00
2026-03	335.36	1.521	149.44	335.20
2026-04	335.36	1.521	149.44	335.19
2026-05	335.36	1.508	149.44	331.84
2026-06	335.36	1.495	149.44	328.54
2026-07	335.36	1.469	149.44	321.97
2026-08	335.36	1.469	149.44	321.98
2026-09	335.36	1.469	149.44	321.98
2026-10	335.36	1.469	149.44	321.98

Appendix B. Model parameters

Table B.8 lists the model parameters. Ranges define triangular distributions whose mode is the central value. Status codes: M measured, O official, O* derived from official data, L literature, Q quotation in the audit report, D design, A assumption, R regulatory scenario. The last nine parameters belong to the hourly net-zero module (Section 3.8) and are varied in scenario analysis only.

Table B.8: Model parameters, ranges and status.

Parameter	Central	Range	Status	Source / note
`discount_rate`	0.1	0.06–0.14	A	student report; sensitivity

Parameter	Central	Range	Status	Source / note
horizon_yr	20	fixed	A	student report
tariff_real_escalation	0	-0.01–0.02	A	constant real tariffs
ef_grid_t_per_MWh	0.444	fixed	O	Secretaría de Medio Ambiente y Recursos Naturales (2025)
gas_share_generation	0.638	0.6–0.68	O	Secretaría de Energía (2025) Tabla 2.2
ccgt_heat_rate_GJ_per_MWh	7.2	6.5–8.5	A	typical CCGT 42-55% LHV
gas_import_share	0.74	0.7–0.78	O*	U.S. Energy Information Administration (2025) 6.4/8.6 Bcf/d
ca_saving_frac	0.2	0.1–0.3	L	U.S. Department of Energy and Compressed Air Challenge (2016) leaks 20-30% -¿ ¡10%; 1%/2 psi
ca_capex	50000	35000–80000	Q	student estimate
ca_om	8000	5000–12000	A	annual ultrasonic leak survey
ca_life	10	fixed	A	
sew_motor_share	0.7	0.5–0.85	A	share of stitching-area energy in clutch motors
sew_saving_frac	0.65	0.5–0.75	L	Nemeša (2023)
sew_motor_kW	0.35	0.25–0.45	L	clutch motor running power
sew_cost_per_unit	3800	2800–5500	A	installed servo motor, market
sew_life	10	fixed	A	
heat_share	0.65	0.5–0.8	A	heater share of vulcanising+injection energy
heat_saving_frac	0.2	0.1–0.3	L	Kent (2018); grey literature 15-30%
heat_capex	100000	60000–150000	A	jackets + controls
heat_life	8	fixed	A	
base_saving_frac	0.1	0.05–0.2	L	Mouzon et al. (2007); Kara and Li (2011)
base_capex	40000	20000–70000	A	timers, contactors, submeters
base_life	10	fixed	A	
pfc_kvar	50	fixed	Q	student design
pfc_pf_after	0.97	0.95–0.99	A	automatic bank
pfc_capex	85000	70000–110000	Q	student estimate
pfc_life	15	fixed	A	

Parameter	Central	Range	Status	Source / note
pv_kWp	163.8	fixed	D	252 x 650 W modules (student design)
pv_cost_per_W	11	9–14	Q	student quote 1.8 MMXN / 163.8 kWp
pv_PR	0.8	0.75–0.85	L	Reich et al. (2012)
pv_tilt_deg	20	fixed	D	latitude tilt
pv_albedo	0.2	fixed	A	
pv_degradation	0.005	0.003–0.008	L	typical crystalline Si
pv_om_frac	0.01	0.005–0.015	A	of capex per year
pv_inverter_repl_frac	0.1	0.08–0.15	A	of capex at year 12
pv_life	25	fixed	L	
pv_m2_per_kWp	5.5	5–6.5	A	650 W module 3.1 m2 incl. spacing
pv_demand_credit	1	0–1	R	net metering reduces Q-based billed demand
lpg_price_MXN_per_kg_incl_vat	19.74	fixed	O	CNE max price, PROFECO 11-17 Oct 2026
lpg_LHV_MJ_per_kg	47.3	fixed	L	IPCC 2006 Vol.2 Table 1.2 NCV
lpg_EF_kg_per_GJ	63.1	fixed	L	IPCC 2006 Vol.2 Table 2.2
ng_EF_kg_per_GJ	56.1	fixed	L	IPCC 2006 Vol.2 Table 2.2
ng_enduser_markup	2	1.5–3	A	transport+distribution over IPGN
boiler_eff	0.85	0.8–0.9	A	
hp_cop	3	2.5–4	L	Arpagaus et al. (2018); Schlosser et al. (2020) (¡100 C)
ca_heat_recovery_frac	0.7	0.5–0.8	L	U.S. Department of Energy and Compressed Air Challenge (2016) ¿80% available
night_load_kW	15	8–25	A	always-on load outside shifts; needs interval data
bat_cost_per_kWh	8500	fixed	A	installed C&I LFP; NREL ATB 2024 (National Renewable Energy Laboratory, 2024) pack 199 USD/kWh + inverter/BOS
bat_rte	0.85	fixed	L	NREL ATB 2024 (National Renewable Energy Laboratory, 2024) commercial battery

Parameter	Central	Range	Status	Source / note
bat_dod	0.9	fixed	A	usable fraction of nameplate
bat_c_rate	0.5	fixed	A	kW per kWh
bat_om_frac	0.025	fixed	L	NREL ATB 2024 (National Renewable Energy Laboratory, 2024): 2.5% of capex per year
bat_life	15	fixed	L	NREL ATB 2024 (National Renewable Energy Laboratory, 2024)
expired_credit_value	0	0–1	A	MXN/kWh paid for credits older than 12 months (PML); 0 = conservative
net_billing_export_value	1	0.6–1.5	A	MXN/kWh export value in a net-billing stress case

Appendix C. Photovoltaic resource and yield

Monthly mean daily plane-of-array irradiation $\bar{H}_T$ was computed from NASA POWER monthly mean global ($\bar{H}$) and diffuse ($\bar{H}_d$) horizontal irradiation (NASA Langley Research Center, 2026) as

$$\bar{H}_T = \bar{H}\left[\left(1 - \frac{\bar{H}_d}{\bar{H}}\right)\bar{R}_b + \frac{\bar{H}_d}{\bar{H}}\,\frac{1+\cos\beta}{2} + \rho_g\,\frac{1-\cos\beta}{2}\right], \qquad \text{(C.1)}$$

with the monthly beam tilt factor $\bar{R}_b$ of Klein (1977) for a south-facing surface evaluated on each month's recommended day, tilt $\beta = 20°$ and ground reflectance $\rho_g = 0.2$. Annual mean horizontal irradiation is 5.05 kWh m^{-2} d^{-1}, compared with the 4.8 peak-sun hours used in the audit report, and plane-of-array irradiation is 5.23 kWh m^{-2} d^{-1}. With a performance ratio of 0.80 the specific yield is 1,526 kWh kWp^{-1} yr^{-1}. The audit report sized the array at 157.8 kWp to cover 80% of a 28,400 kWh month with 4.8 peak-sun hours and no performance ratio, and proposed 252 modules (163.8 kWp). This study adopts the module-based capacity.

Appendix D. Sensitivity analysis

Table D.9 gives the 20-year NPV of each portfolio with each parameter at its low and high bound and all others at their central values, ranked by the swing of P4.

Table D.9: One-way sensitivity of 20-year NPV (MMXN).

	P1 Efficiency		P2 PV only		P4 Efficiency first	
Parameter	low	high	low	high	low	high
discount_rate	2.64	1.33	5.42	2.39	5.64	2.64
tariff_real_escalation	1.68	2.22	3.23	4.34	3.45	4.62
base_saving_frac	1.56	2.40	3.57	3.57	3.53	4.36
pv_cost_per_W	1.84	1.84	3.92	3.03	4.01	3.50
heat_saving_frac	1.60	2.08	3.57	3.57	3.57	4.04
pv_PR	1.84	1.84	3.22	3.91	3.61	4.00
sew_saving_frac	1.64	1.97	3.57	3.57	3.61	3.94
ca_saving_frac	1.68	2.00	3.57	3.57	3.65	3.96
sew_motor_share	1.69	1.95	3.57	3.57	3.66	3.92
sew_cost_per_unit	1.93	1.69	3.57	3.57	3.89	3.66
heat_share	1.73	1.95	3.57	3.57	3.70	3.91
sew_motor_kW	1.71	1.91	3.57	3.57	3.67	3.88

Appendix E. Measurement and verification plan

The audit team prepared an M&V plan following IPMVP (Efficiency Valuation Organization, 2022). PV output is verified under Option B through inverter and bidirectional-meter data with local irradiance. The capacitor bank is verified under Option C through billed PF and surcharges, or Option B where a network analyser is installed. Compressed air is verified under Option B through dedicated metering of kWh, pressure and operating hours, with specific consumption (kWh per m^3 of air) as the indicator where flow can be measured. Servo motors and insulation can be verified under Option A with spot power measurements and logged hours. Base-load control is verified under Option C through the production-normalised baseline of Section 3, complemented by night and weekend load readings. The whole facility is verified under Option C with at least 12 months of post-implementation billing, adjusted for production and operating hours. Given the CV-RMSE of the baseline (16.0%), savings below about 10% of consumption cannot be detected reliably at whole-facility level. Sub-metering of the compressed-air room and of the vulcanising and stitching areas is therefore recommended as the first implementation step.

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